

REVIEW OF (LAST PART OF) LECTURE 4

- Joint pmf of X, Y : $f(x, y) = P(X = x \text{ and } Y = y)$
- Joint pdf of X, Y : $f(x, y) \geq 0$ AND $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$

$$P((X, Y) \in C) = \int_C \int f(x, y) \, dx \, dy$$

- Marginal pmf of discrete X : $f_1(x) = \sum_{\forall y} f(x, y)$
- Marginal pdf of continuous X : $f_1(x) = \int_{-\infty}^{\infty} f(x, y) \, dy$

INDEPENDENT RANDOM VARIABLES

- X, Y are independent (denoted with $X \perp\!\!\!\perp Y$) if and only if

$$f(x, y) = f_1(x)f_2(y) \quad \forall x, y$$

Ex: Which of these corresponds to independent X, Y ?

	$Y = 0$	$Y = 1$	
$X=0$	$1/4$	$1/4$	$1/2$
$X=1$	$1/4$	$1/4$	$1/2$
	$1/2$	$1/2$	1

	$Y = 0$	$Y = 1$	
$X=0$	$1/2$	0	$1/2$
$X=1$	0	$1/2$	$1/2$
	$1/2$	$1/2$	1

INDEPENDENT RANDOM VARIABLES

- X, Y are independent (denoted with $X \perp\!\!\!\perp Y$) if and only if

$$f(x, y) = f_1(x)f_2(y) \quad \forall x, y$$

Ex: Which of these corresponds to independent X, Y ?

	$Y = 0$	$Y = 1$	
$X=0$	$1/4$	$1/4$	$1/2$
$X=1$	$1/4$	$1/4$	$1/2$
	$1/2$	$1/2$	1

X, Y independent.

	$Y = 0$	$Y = 1$	
$X=0$	$1/2$	0	$1/2$
$X=1$	0	$1/2$	$1/2$
	$1/2$	$1/2$	1

X, Y not independent.

$$f_1(0)f_2(1) = \frac{1}{4} \text{ whereas } f(0,1) = 0$$

CONDITIONAL DISTRIBUTIONS

- **Discrete:** X, Y have joint pmf f . Y has marginal pmf f_2 .

Conditional pmf of X given Y : $g_1(x | y) = \frac{f(x, y)}{f_2(y)}$

Note: $P(X = x | Y = y) = g_1(x | y)$

CONDITIONAL DISTRIBUTIONS

- **Discrete:** X, Y have joint pmf f . Y has marginal pmf f_2 .

Conditional pmf of X given Y : $g_1(x | y) = \frac{f(x, y)}{f_2(y)}$

Note: $P(X = x | Y = y) = g_1(x | y)$

- **Continuous:** X, Y have joint pdf f . Y has marginal pdf f_2 .

Conditional pdf of X given Y : $g_1(x | y) = \frac{f(x, y)}{f_2(y)}$

Note: $P(X \in C | Y = y) = \int_C g_1(x | y) dx$

DISCRETE CONDITIONAL DISTRIBUTIONS

Ex: $X = 0$: Car not stolen, $X = 1$: Car stolen

The table corresponding to joint pmf of X, Y :

Stolen X	Brand Y					Total
	1	2	3	4	5	
0	0.129	0.298	0.161	0.280	0.108	0.976
1	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

Give the table corresponding to conditional pmf of X given Y .

DISCRETE CONDITIONAL DISTRIBUTIONS

Ex: $X = 0$: Car not stolen, $X = 1$: Car stolen

The table corresponding to joint pmf of X, Y :

Stolen X	Brand Y					Total
	1	2	3	4	5	
0	0.129	0.298	0.161	0.280	0.108	0.976
1	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

Give the table corresponding to conditional pmf of X given Y .

Stolen X	Brand Y				
	1	2	3	4	5
0	0.928	0.968	0.994	0.993	0.991
1	0.072	0.032	0.006	0.007	0.009

CONTINUOUS CONDITIONAL DISTRIBUTIONS

Ex: Y : time required to do necessary reading

X : time required to do the homework (includes Y)

$$f(x, y) = \begin{cases} e^{-x} & \text{for } 0 \leq y \leq x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find $P(X \geq 10 \mid Y = 4)$?

Useful integral:

$$\int_a^\infty e^{-x} = e^{-a}$$

CONTINUOUS CONDITIONAL DISTRIBUTIONS

Ex: Y : time required to do necessary reading

X : time required to do the homework (includes Y)

Useful integral:
 $\int_a^\infty e^{-x} = e^{-a}$

$$f(x, y) = \begin{cases} e^{-x} & \text{for } 0 \leq y \leq x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find $P(X \geq 10 \mid Y = 4)$?

marginal pmf of Y : $f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_y^{\infty} e^{-x} dx = e^{-y}$

$$g_1(x \mid y) = \frac{f(x, y)}{f_2(y)} = \frac{e^{-x}}{e^{-y}} = e^{y-x} \Rightarrow P(X \geq 10 \mid Y = 4) = \int_{10}^{\infty} e^{4-x} dx = e^{-6}$$

GENERALIZING REST OF THE RULES ON EVENTS

- **Multiplication Rule:** $f(x, y) = g_1(x | y)f_2(y)$
- **Law of Total Probability:**

$$f_1(x) = \sum_y g_1(x | y)f_2(y) \quad \text{for discrete } Y$$

$$f_1(x) = \int_{-\infty}^{\infty} g_1(x | y)f_2(y) dy \quad \text{for continuous } Y$$

- **Bayes' Theorem :**

$$g_1(x | y) = \frac{g_2(y | x)f_1(x)}{f_2(y)}$$

EXPECTATION

- Readings for the next few lectures:

Ch 8 (Watkins)

Ch 3 (Wasserman, for more formal treatment)

EXPECTATION

- Mean $\equiv$ expected value $\equiv$ expectation
- Weighted average of values of a random variable where weights are probabilities.

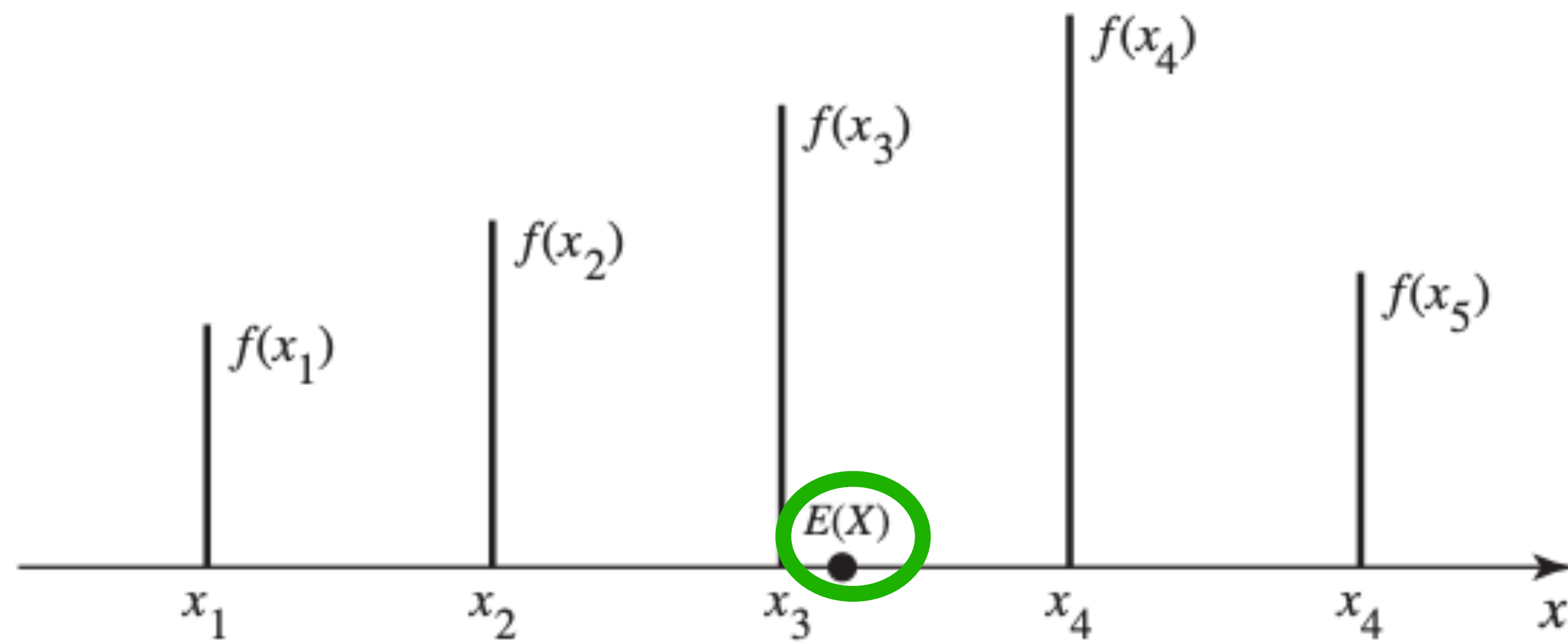
$$E(X) = \sum_{\forall x} xf(x) \quad \text{for discrete } X$$

$$E(X) = \int xf(x) \, dx \quad \text{for continuous } X$$

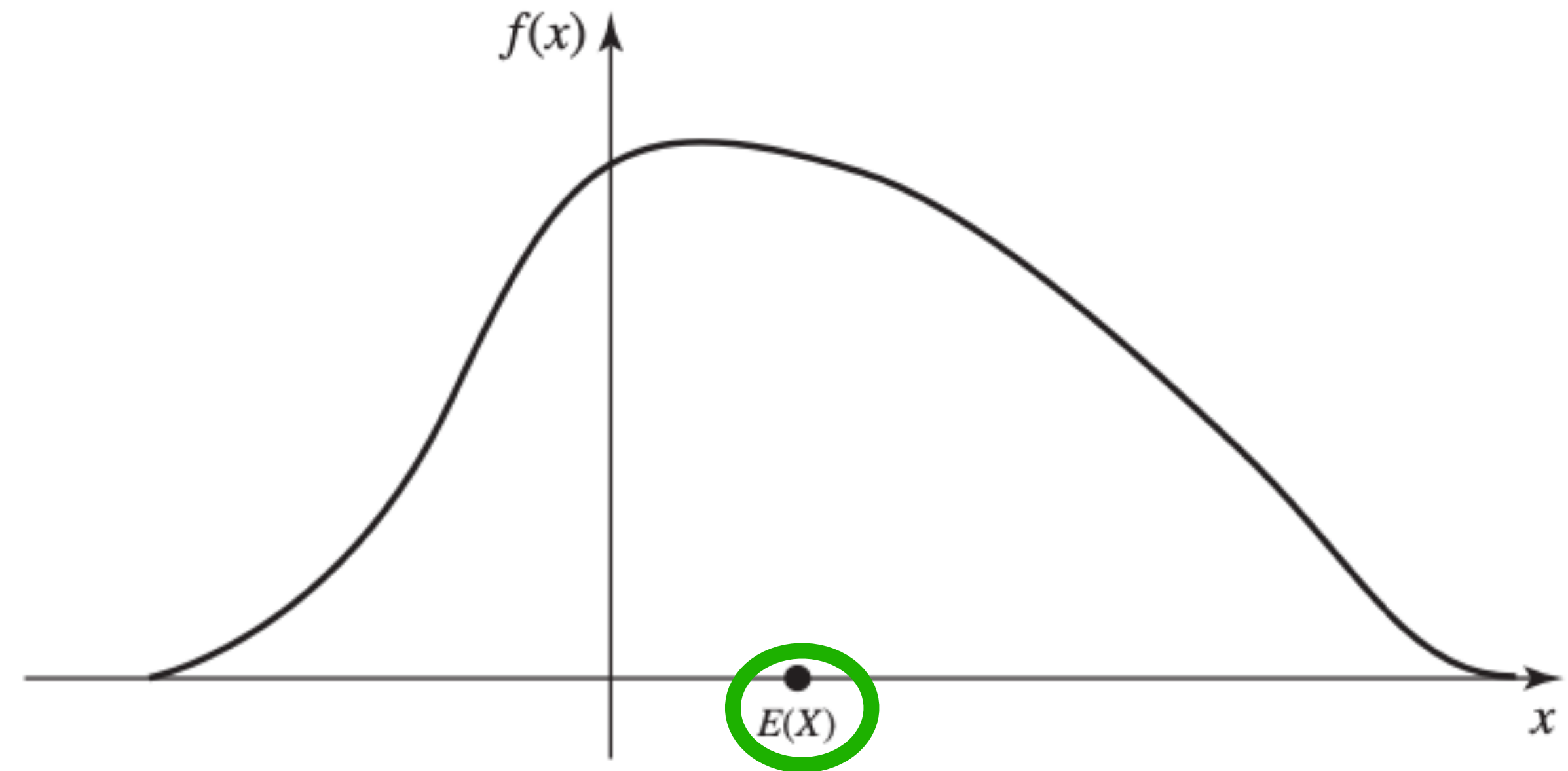
- Also denoted with μ .

EXPECTATION

- Expectation as the center of gravity



Discrete

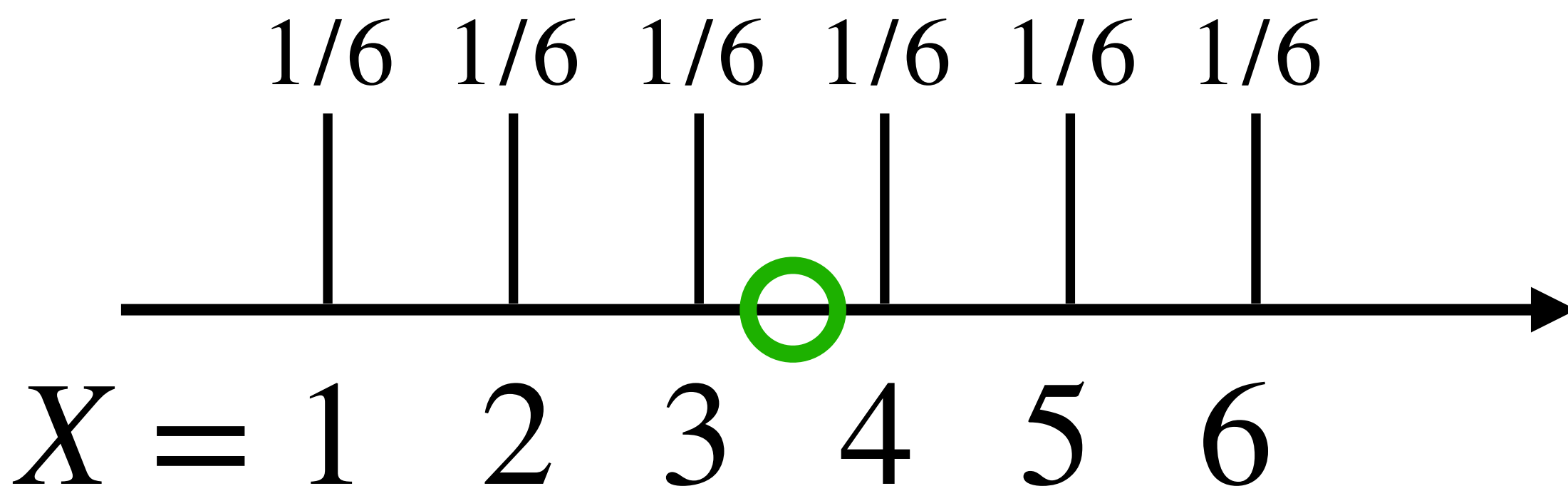


Continuous

EXPECTATION

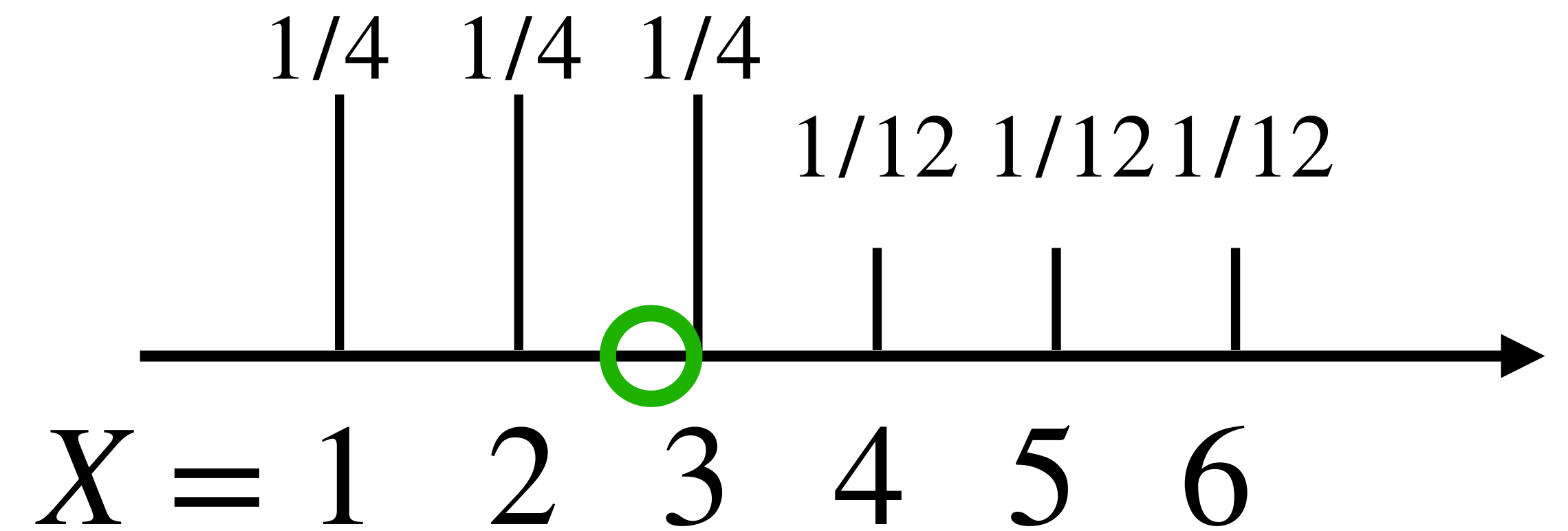
Ex: Expectation of fair die vs unfair die. X : value on die.

Fair die



$$E(X) = 1 \times 1/6 + 2 \times 1/6 + \dots + 6 \times 1/6 = 7/2$$

Unfair die



$$E(X) = 1 \times 1/4 + 2 \times 1/4 + \dots + 6 \times 1/12 = 11/4$$

EXPECTATION

Ex: Expectation of Bernoulli random variable X

X has Bernoulli distribution with parameter $p \Rightarrow$
 $P(X = 1) = p, P(X = 0) = 1 - p$

EXPECTATION

Ex: Expectation of Bernoulli random variable X

X has Bernoulli distribution with parameter $p \Rightarrow$
 $P(X = 1) = p, P(X = 0) = 1 - p$

- Throwing a coin with probability of Head p , Tail $1 - p$

EXPECTATION

Ex: Expectation of Bernoulli random variable X

X has Bernoulli distribution with parameter $p \Rightarrow$
 $P(X = 1) = p, P(X = 0) = 1 - p$

- Throwing a coin with probability of Head p , Tail $1 - p$
- Sampling with replacement: **R**, **B** balls in a box.
Proportion of **R** is p . $X_i = 1$, if i^{th} ball is **R**, 0 otherwise.

EXPECTATION

Ex: Expectation of Bernoulli random variable X

X has Bernoulli distribution with parameter $p \Rightarrow$
 $P(X = 1) = p, P(X = 0) = 1 - p$

- Throwing a coin with probability of Head p , Tail $1 - p$
- Sampling with replacement: **R**, **B** balls in a box.
Proportion of **R** is p . $X_i = 1$, if i^{th} ball is **R**, 0 otherwise.

What is $E(X)$?

EXPECTATION

Ex: Expectation of Bernoulli random variable X

X has Bernoulli distribution with parameter $p \Rightarrow$
 $P(X = 1) = p, P(X = 0) = 1 - p$

- Throwing a coin with probability of Head p , Tail $1 - p$
- Sampling with replacement: **R**, **B** balls in a box.
Proportion of **R** is p . $X_i = 1$, if i^{th} ball is **R**, 0 otherwise.

What is $E(X)$?

$$E(X) = \sum_{x=\{0,1\}} xf(x) = 0 \times (1 - p) + 1 \times p = p$$

EXPECTATION

Ex: X : Time until a lightbulb fails. Its pdf $f(x) = 2x$, $0 < x < 1$.

What is $E(X)$?

EXPECTATION

Ex: X : Time until a lightbulb fails. Its pdf $f(x) = 2x$, $0 < x < 1$.

What is $E(X)$?

$$E(X) = \int x f(x) \, dx = \int_0^1 x(2x) \, dx = \int_0^1 2x^2 \, dx = \frac{2}{3}$$

EXPECTATION OF A FUNCTION OF A RANDOM VARIABLE

- For continuous random variable X , let $r(X)$: a function of X

$$E[r(X)] = \int r(x)f(x) dx$$

(Rule of the lazy statistician: could also find it by first finding pdf of $r(X)$ which would require many further calculations. Lazy prefers easy.)

- Same for discrete RV, except replace integral with sum.

EXPECTATION OF A FUNCTION OF A RANDOM VARIABLE

Ex: Assume pdf of previous example, i.e. $f(x) = 2x$, $0 < x < 1$

Find $E(\sqrt{X})$.

$$E(\sqrt{X}) = \int \sqrt{x} f(x) dx = \int_0^1 \sqrt{x} 2x dx = \int_0^1 2x^{3/2} dx = \frac{4}{5}$$

FURTHER PROPERTIES

- Sums: Expectation of sum is sum of expectations

For random variables X_1, X_2

$$E(X_1 + X_2) = E(X_1) + E(X_2)$$

Note: Generalizes to n variables.

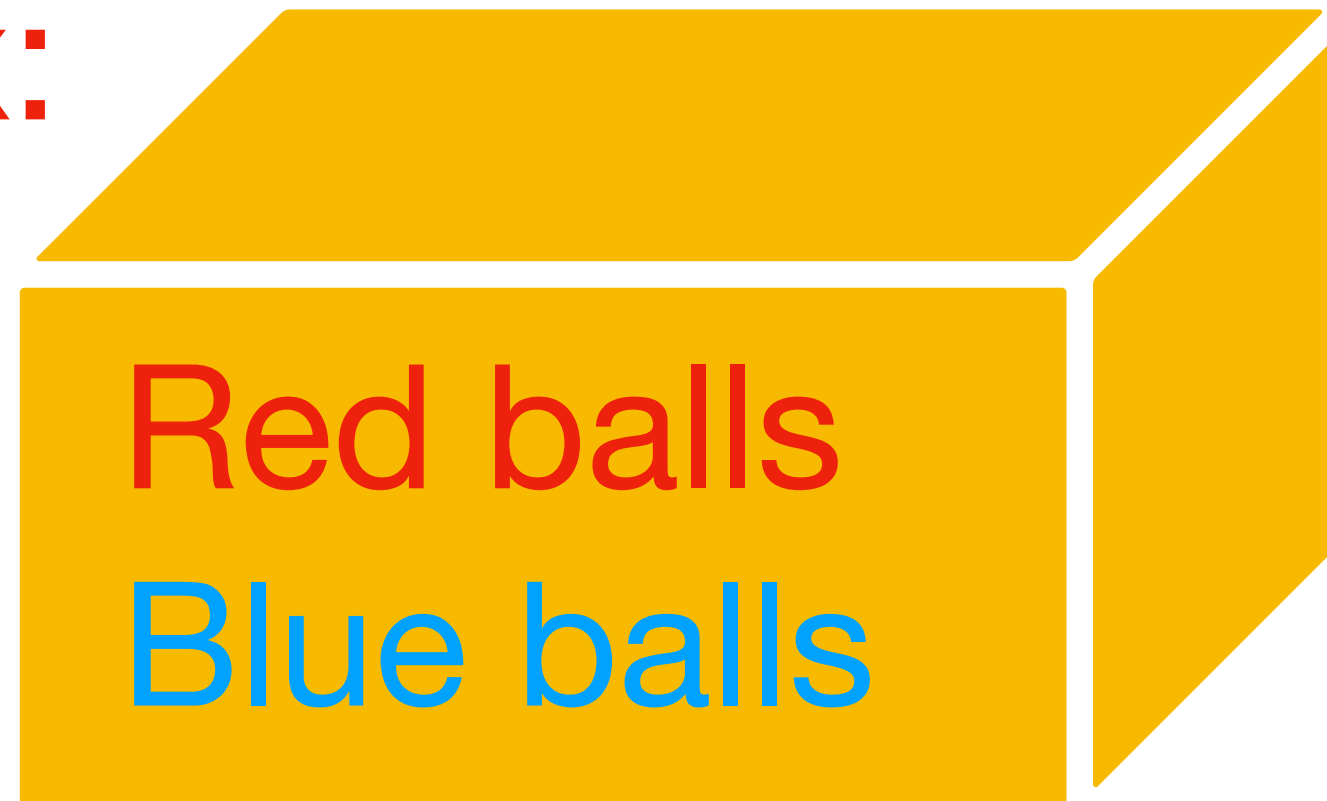
- Also implies linearity of expectation

Let $Y = aX + b$ for constants a, b

$$E(Y) = aE(X) + b$$

FURTHER PROPERTIES

Ex:

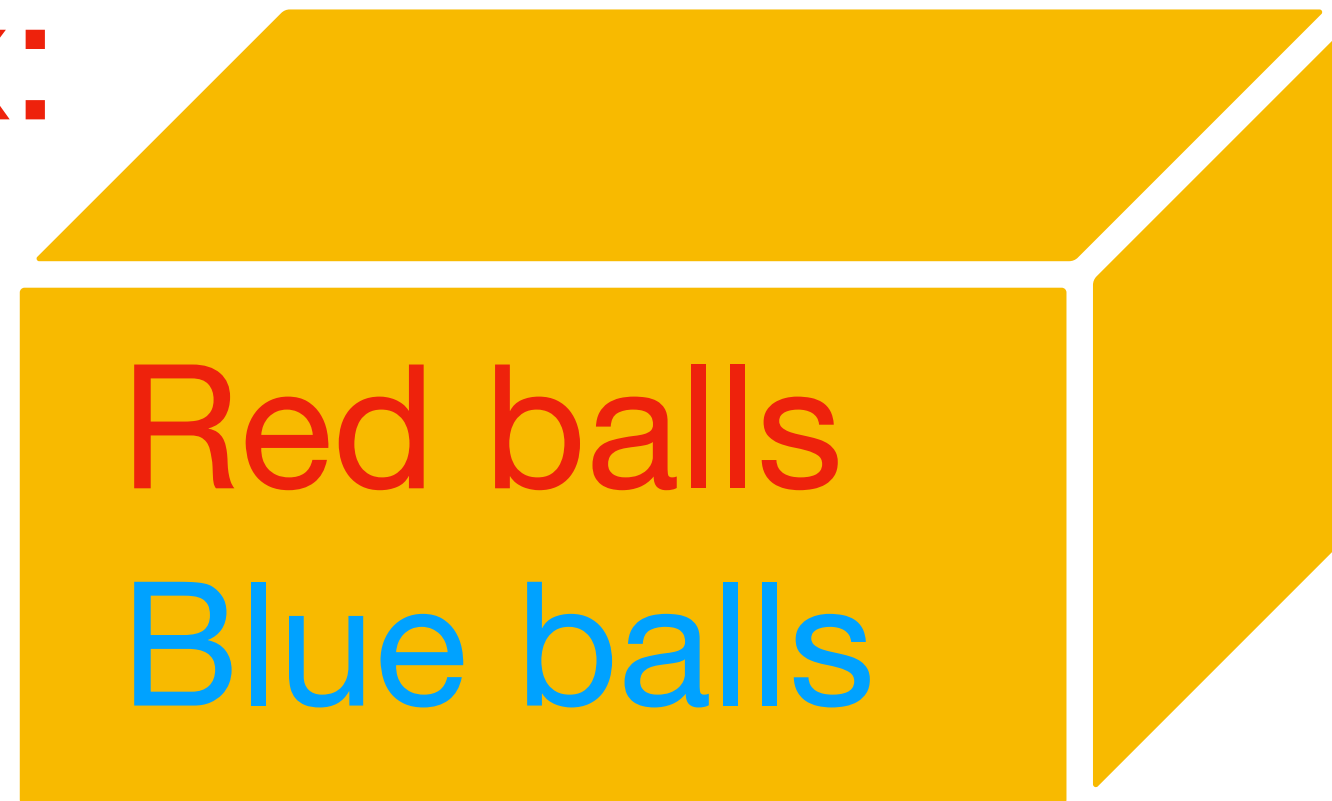


Proportion of R is p . Random sample of n balls selected **with replacement**.

X : # of R balls in the sample. $E(X) = ?$

FURTHER PROPERTIES

Ex:



Proportion of R is p . Random sample of n balls selected **with replacement**.

X : # of R balls in the sample. $E(X) = ?$

Let $X_i = 1$ if i^{th} ball is red, 0 otherwise (for $i = 1, \dots, n$).

$$\Rightarrow X = X_1 + X_2 + \dots + X_n$$

X_i has **Bernoulli distribution** with parameter p .

$$\Rightarrow E(X) = E(X_1 + \dots + X_n) = E(X_1) + \dots + E(X_n) = np$$

CONDITIONAL EXPECTATION

- Conditional expectation of X given $Y = y$:

$$E(X | Y = y) = \sum_x x g_1(x | y) \quad (\text{if } X \text{ has discr. cond. distr. given } Y = y)$$

$$E(X | Y = y) = \int_{-\infty}^{\infty} x g_1(x | y) \quad (\text{if } X \text{ has cont. cond. distr. given } Y = y)$$

CONDITIONAL EXPECTATION

Ex: Roll 2 fair dice. Expected value of die 1 given their sum is 5?

X : outcome of die 1, Y : Sum of 2 dice

$$E(X | Y = y) = \sum_x x g_1(x | y) = \sum_x x \frac{f(x, y)}{f_2(y)}$$

$$E(X | Y = 5) = \sum_{x=1}^{x=4} x \frac{1/36}{4/36} = \sum_{x=1}^{x=4} x \frac{1}{4} = 2.5$$